TPC-MVP1: A Falsifiable Proof Architecture for Fixed-Shift Prime-Pair Correlations:

Verified Interfaces, Explicit Working Hypotheses, and Stop–Go Tests at the Parity Frontier

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Abstract

The first 102 papers of the TPC fixed-shift program contain exact identities, literal carrier dictionaries, finite spectral theorems, selected packet closures, obstruction results, and several dependency audits. They do not contain a fixed-shift prime-pair theorem. This paper deliberately changes scale: it compresses that infrastructure into a single falsifiable proof architecture.

We first record a coarse seven-stage audit and isolate the results that may be imported without conjecture. These include the reduction of a weighted fixed-shift prime correlation to one named hard packet, the determinant and full-block interfaces, lossless low-window affine export, endpoint-scale return of two literal exceptional resonance branches within the present low-window decomposition, an exact multiplicative resonance spectrum, and a width-sensitive three-ledger inequality for the remaining positive resonant carrier.

We then formulate three macro hypotheses—literal completeness, literal packet saving, and physical synthesis—resolved into nine separately auditable gates. One gate, a growing fixed-h_0 signed affine Möbius estimate, is explicitly identified as the parity-facing arithmetic hypothesis; it is neither proved nor called technical. Under the nine hypotheses, with all objects in one normalization and a total physical loss strictly below $1/400$, we prove a conditional weighted Hardy–Littlewood asymptotic. For a nonnegative weight this conditionally gives $\gg X/\log^2 X$ prime pairs in a dyadic interval, and only then may one specialize to $h_0 = 2$.

The contribution is therefore a map rather than an exit from the maze: every conjectural edge has a go certificate, a genuine stop certificate, and a publishable fallback. No new L2 estimate, parity breakthrough, twin-prime conclusion, Riemann-hypothesis consequence, or spectral identification of zeta zeros is asserted.

Keywords: fixed-shift prime pairs; parity barrier; Möbius correlation; determinant dispersion; literal arithmetic carrier; conditional synthesis; falsifiable roadmap.

Status statement. The TPC series has established many exact algebraic, spectral, provenance, and literal physical-interface results, but no growing fixed-h_0 arithmetic cancellation theorem. TPC-MVP1 is a *minimum viable proof program*: a falsifiable conditional architecture, not a proof of a prime-pair theorem and not a claim that the remaining hypotheses are merely technical.

1 Purpose, claim taxonomy, and the fixed-shift target

1.1 Four claim tags and three proof levels

Every major statement in this paper carries one of four logical tags:

[E] established, **[H]** working hypothesis, **[C]** conditional synthesis, **[N]** not established.

The tags answer a different question from the inherited proof levels: L0 denotes finite algebra, exact model identities, abstract operator theorems, or countermodels; L1 denotes a literal physical-frame interface preserving the actual arithmetic typing; and L2 is reserved for a new growing estimate on the complete coefficient attached to one prescribed nonzero shift h_0.

Definition 1.1 (Promotion rule)**.** An averaged-shift, almost-all-scale, frozen-family, random-sign, support-only, positive-square, or surrogate-carrier estimate is not an L2 estimate for the physical fixed-h_0 coefficient unless an explicit theorem returns it to that coefficient with all losses and exceptional sectors controlled.

This paper is an L1 synthesis. A conditional implication may have an L2 conclusion while none of its unproved hypotheses is promoted to an established L2 theorem.

1.2 The target correlation

Fix throughout a nonzero even integer h_0, and let $W \in C_c^\infty((0,\infty))$ be fixed. Write

$$\mathcal{C}_{h_0,W}(X) = \sum_{n\geq 1} \Lambda(n)\Lambda(n+h_0)W(n/X).$$

If $\nu_{h_0}(p)$ is the number of residue classes occupied by $\{0,-h_0\} \pmod p$, define the usual singular series

$$\mathfrak{S}(h_0) = \prod_p \frac{1-\nu_{h_0}(p)/p}{(1-1/p)^2}.$$

It is positive for every fixed admissible even h_0. The centered defect is the classical prime-pair target [1, 2]:

$$\Delta_{h_0,W}(X) = \mathcal{C}_{h_0,W}(X) - \mathfrak{S}(h_0)X\int_0^\infty W(t)\,dt. \tag{1}$$

Proposition 1.2 ([E] Established hard-packet anchor)**.** *The TPC fixed-shift local-model reduction gives, for a suitable fixed* $0<\delta<1/2$*,*

$$\Delta_{h_0,W}(X) = B_{h_0,\delta}(X) + O_{A,h_0,W,\delta}\Big(X(\log X)^{-A}\Big) \tag{2}$$

for every fixed $A>0$*, where* $B_{h_0,\delta}$ *is an explicitly defined bilinear hard packet. Consequently the weighted Hardy–Littlewood asymptotic is equivalent, within that reduction, to*

$$B_{h_0,\delta}(X) = o(X).$$

This is the entrance to the architecture, not an estimate of the hard packet. The purpose of the later TPC layers is to expose its literal geometry without replacing it by an easier random or averaged object [7, 8].

1.3 Six invariants that no route may drop

Every hypothesis and every future promotion is audited against:

- **I1.** the literal physical coefficients, including their signs and reconstruction multipliers;
- **I2.** one prescribed nonzero h_0, not an averaged or subsequently chosen shift;
- **I3.** physical atomic normalization and exactly one global normalization;
- **I4.** canonical or losslessly invertible native keys, without null-pair inflation;
- **I5.** the actual active support, masks, prefixes, polarizations, short fibers, and boundary sectors; and
- **I6.** a nonduplicated endpoint ledger whose total physical loss is strictly below $1/400$.

Failure of any item prevents a claim of final proof progress. In particular, equality at the physical endpoint is a stop.

2 What 102 papers actually established

The papers TPC-1 through TPC-102 are consecutive, but they are not 102 monotone steps toward a proof. They include exact identities, finite spectral calculations, literal dictionaries, conditional interfaces, selected packet theorems, sharp countermodels, and route audits. A useful compression is the following seven-stage ledger.

Stage	Papers	Established content at coarse resolution	Honest status
A	TPC-1–9	Two-scale sieve correlations, singular-series calibration, prime-target defect identities, and the exact appearance of the large-divisor fixed-shift tail. Ordinary spectral flatness is shown not to control rare prime-pair events.	L0/L1 entrance and no-go
B	TPC-10–18	Factor-scale and fixed-shift Type-II normal forms, the named hard packet, two TT^* quadraticizations, determinant variables, exact endpoint collapses, and a conditional full-block closure template.	L1 reduction; determinant dispersion open
C	TPC-19–36	Primitive determinants, conductor and fiber decompositions, dynamic cutoffs, one-sided Poisson passages, joint-before-centering rules, selected calibrated squares, and the distinguished zero-frequency gate.	Several selected L1 blocks; full tail open
D	TPC-37–55	CRT and alias tomography, native Gram expansions, four-Möbius isolation, partial sign freezing, all-band tiling, atomic distortion, literal cell survival, and canonical-frame obstructions.	Physical interfaces plus sharp no-go theorems
E	TPC-56–72	Shorted-Gram reassembly, exact quantifier-safe closure alternatives, native affine provenance, residual grouping, primitive cofactor planes, trace/operator nonteleportation, and a ten-paper route audit.	L0/L1 carrier architecture
F	TPC-73–92	Primitive support and completion certificates, actual-support incidence, scalar holonomy, determinant-energy transfer, zero-mode barriers, Tauberian defects, local squarefree support, and exact full-block cover criteria.	Five separately required gates; no L2 theorem
G	TPC-93–102	Lossless literal low-window affine export, exact conductor and progression ledgers, anchored translation formulas, safe zero-mode projection, two exceptional branch returns, exact multiplicative resonance spectrum, opened-atom linearization, and the three-ledger master inequality.	Latest L1 resonance route

Table 1: A coarse status ledger for TPC-1 through TPC-102.

Representative anchors for the seven stages are TPC-1 and TPC-15 for A, TPC-18 for B, TPC-28 and TPC-32 for C, TPC-50 and TPC-51 for D, TPC-70 for E, TPC-82 and TPC-92 for F, and TPC-102 for G [3, 6–13, 15, 19].

2.1 What moved forward

The principal progress is not a numerical count of manuscripts. Four qualitative changes are durable.

1. The target was reduced from an informal parity narrative to the named scalar defect $B_{h_0,\delta}(X)$ in (2).

2. Many changes of variables are now losslessly typed: source, content, conductor, determinant, affine slope, mask, polarization, and global normalization can be traced in important low-window and residual sectors [8, 13, 20].

3. Several seductive but invalid shortcuts have exact countermodels: support or unsigned energy does not determine the zero mode; continuous signed Mellin representations do not possess a uniform finite lower frame; logarithmic cancellation does not automatically translate to ordinary cancellation; and a singleton cell does not provide cross-cell dispersion [4, 12, 14, 17].

4. The latest resonance carrier is no longer a vague spectral object. Its constant and $q_X \mid u$ exceptional branches are separately returned within the present low-window decomposition and native normalization, and its remaining positive part is governed by three explicit weighted ledgers [5, 6, 22].

2.2 What did not move

No paper through TPC-102 proves a growing estimate for the complete literal coefficient attached to one prescribed h_0. In particular:

- no fixed-shift Möbius or Liouville cancellation theorem is available on the full generic affine carrier;
- no determinant-energy lower bound and distinguished zero-mode estimate have been proved simultaneously on the actual support;
- no all-block physical return with strict endpoint slack is complete; and
- no parity breakthrough, prime-pair lower bound, or twin-prime theorem has been obtained.

The dependency graph has become narrower and more falsifiable; no quantitative distance to a proof is asserted.

3 The established ledger

This section collects only facts that the MVP architecture imports as proved. Their conjunction still falls short of an L2 estimate.

Proposition 3.1 ([E] E1: exact target anchoring)**.** *The target defect is anchored by* (2)*. Thus a proof of $B_{h_0,\delta}(X) = o(X)$, with the actual fixed-h_0 coefficient, is sufficient and necessary within the imported local-model reduction.*

Proposition 3.2 ([E] E2: determinant and full-block interfaces)**.** *The Möbius tail admits exact TT^*, lcm-return, content, determinant, and primitive/nonprimitive decompositions. A generic primitive determinant-dispersion input together with an all-block cover would conditionally close the tail. Neither that dispersion input nor the all-block cover is asserted as proved [8, 18].*

Proposition 3.3 ([E] E3: literal carrier typing)**.** *On the sectors for which a literal dictionary has been certified, the physical coefficient can be transported while retaining h_0, source keys, content/projector children, both polarizations, masks, prefixes, Möbius signs, actual support, and one global normalization. TPC-93 gives a lossless low-window affine export and restores the reduced determinant to the physical value h_0 [20].*

Proposition 3.4 ([E] E4: route exclusions)**.** *The following implications are false without additional literal input:*

(i) ordinary Haar spectral behavior implies rare prime-pair correlation [3];

(ii) support, completion, or unsigned energy implies zero-mode cancellation [14, 15];

(iii) arbitrary continuous signed Mellin synthesis has a uniform finite lower frame [12];

(iv) anchor-free logarithmic cancellation implies translated ordinary cancellation [16, 21];

(v) squarefree local density implies Möbius-sign cancellation [17]; and

(vi) conductor q_X, singleton fine cells, or bounded overlap by itself implies physical resonance dispersion or full-block cover [4, 18, 23].

Proposition 3.5 ([E] E5: two literal exceptional returns)**.** *Within the present literal low-window decomposition, the conductor-one constant-phase branch and the nonconstant $q_X \mid u$ opened branch are each returned at the recorded endpoint scale*

$$X^{o(1)}Q_X^2.$$

These are L1 *branch closures. They are not estimates of the complete signed low-window coefficient [4, 22].*

Theorem 3.6 ([E] E6: three-ledger resonance inequality)**.** *For the remaining positive invertible resonant carrier, define*

$$\mathfrak{P}_X = \sum_{H\in\mathcal{H}_X} \frac{2H}{q_X-1} M_H^{\mathrm{inv}}, \qquad \mathfrak{X}_X = \sum_{H\in\mathcal{H}_X} g_{q_X}(H)\sqrt{|\mathfrak{X}_H^{\circ}|},$$

where

$$g_q(H) = \sqrt{\frac{2H(q-1-2H)}{q-1}},$$

and let W_X be the maximum normalized opened-atom weight. The imported literal results give

$$\mathcal{C}^{\mathrm{abs}}_{\mathrm{exc},X} \ll X^{o(1)}Q_X^2 + B_X\left[\mathfrak{P}_X + \sqrt{W_X|\mathcal{H}_X|(q_X-1)\mathfrak{P}_X} + \mathfrak{X}_X\right]. \tag{3}$$

The first term includes the exceptional returns in proposition 3.5; the three quantities in brackets are not bounded at the required growing scale by the established theory [5, 6].

Proposition 3.7 ([E] E7: two independent exponent ledgers)**.** *If the determinant energy and distinguished zero mode are written as*

$$D_X \geq X^{-\lambda_D+o(1)}Q_X^3J_X^2, \qquad |Z_X| \leq X^{-\eta_Z+o(1)}Q_X^2,$$

then the recorded sufficient compatibility condition is

$$\lambda_D \leq 2\eta_Z. \tag{4}$$

This is independent of the physical synthesis requirement

$$\Lambda_{\mathrm{phys}} < \frac{1}{400}. \tag{5}$$

Equality is compatible in (4) but leaves no determinant reserve; equality in (5) is a stop. The same loss may not be charged to both ledgers [14, 15, 19].

Proposition 3.8 ([E] E8: cover and arithmetic dispersion are independent)**.** *An exact or controlled-remainder full-block cover does not imply primitive determinant cancellation, and a determinant estimate on a selected block does not imply literal full-block reassembly. Both must be supplied in one carrier and one quantifier scope [18, 19].*

Remark 3.9 (The first genuinely new L2 event)*.* A future theorem would count as the first L2 advance only if it reduces a fixed polynomial exponent for the actual growing fixed-h_0 coefficient, uniformly returns every sector required by its statement, and survives the physical normalization. A new exact spectrum or finite certificate, however useful, remains L0/L1.

4 The minimum viable hypothesis stack

Set

$$\sigma_* = \frac{1}{400}.$$

The architecture has three macro hypotheses:

$$\boxed{\textbf{L} \text{ literal completeness}} \;+\; \boxed{\textbf{P} \text{ literal packet saving}} \;+\; \boxed{\textbf{S} \text{ physical synthesis}}$$
$$\Longrightarrow \quad B_{h_0,\delta}(X) = o(X).$$

They are resolved below into nine gates. The resolution is intentionally redundant enough to be falsifiable: a failed gate names the object and the route that failed.

4.1 Macro hypothesis L: literal completeness

Working Hypothesis 4.1 ([H] H1: canonical physical synthesis)**.** *There is a provenance-preserving canonical packet family $\{\mathcal{P}_b\}_{b\in\mathcal{B}_X}$, with $|\mathcal{B}_X| = X^{o(1)}$, defined from the actual fixed-h_0 coefficient. On the actual support, this family is an exhaustive disjoint classification into exceptional resonant, remaining resonant, generic, and returned-complement packets; no physical block lies outside these classes. Its keys retain every invariant in section 1.3. The synthesis map is the inherited physical map, not an arbitrary signed representation, and its cost is entered once in Λ_{phys}.*

Working Hypothesis 4.2 ([H] H4: high-frequency and boundary return)**.** *The high-frequency reconstruction tail, ultra-long residual complement, short and bounded fibers, slope collisions, both polarizations, endpoint bands, and outer keys are either assigned to a soft remainder whose* complete outer sum *is $o(X)$ in the original normalization, or retained as packet terms that satisfy the same raw saving $X^{1-\sigma_*+o(1)}$ used below. Write $\mathcal{B}_X^{\mathrm{ns}} \subseteq \mathcal{B}_X$ for the indices of these retained nonsoft packets. No favorable sector is silently promoted to the full coefficient.*

Working Hypothesis 4.3 ([H] H6: full-block physical return)**.** *The TPC-18 block family covers the literal physical tail in one carrier and one quantifier scope. Either an exact certificate $Bc = w$ holds, or the uncovered physical remainder is estimated separately by $o(X)$. Overlap, missing blocks, and normalization are included in the synthesis operator.*

Working Hypothesis 4.4 ([H] H7: prescribed-shift localization)**.** *Every frozen, aliased, frequency-averaged, or auxiliary-parameter estimate used by the route returns, with subcritical recorded cost, to the same prescribed nonzero h_0. An almost-all or averaged-shift statement does not satisfy this hypothesis.*

Working Hypothesis 4.5 ([H] H8: literal tail reconnection)**.** *After all packet estimates and reassembly, the output intertwines exactly with the original hard packet $B_{h_0,\delta}(X)$ in (2). There is no second global normalization, omitted branch, duplicated atom, or substitution of a surrogate Gram matrix for the native physical Gram matrix.*

Together, H1, H4, H6, H7, and H8 constitute macro hypothesis **L**. They are structural and analytic, but they are not declared easy.

4.2 Macro hypothesis P: packet saving

Working Hypothesis 4.6 ([H] H2: literal positive resonance)**.** *On the actual growing TPC-102 invertible carrier,*

$$B_X \le X^{o(1)}, \qquad W_X \le X^{o(1)}, \qquad \mathfrak{P}_X \le X^{o(1)} Q_X^2, \qquad \mathfrak{X}_X \le X^{o(1)} Q_X^2. \tag{6}$$

These bounds use the opened atomic weights and active widths, not an ambient branch census. The exact inherited normalization transports (3), including its leading exceptional term and the two E5 returns, to the raw packet target

$$|\mathcal{P}_{b,\mathrm{res}}| \le X^{1-\sigma_*+o(1)}.$$

The last sentence is part of the hypothesis because (3) is a native TT^*-scale statement. A normalization crosswalk must be proved rather than inferred by dimension counting.

Working Hypothesis 4.7 ([H] H3: restricted generic affine arithmetic)**.** *For every actual long generic affine block, retain in the same sum the literal Möbius weights, the r-dependent phase, the affine character, the reconstruction multiplier, all prefixes, and the fixed h_0. Its normalized signed TT^* correlation satisfies the working target*

$$\mathcal{G}_{b,\mathrm{gen}}^{\mathrm{signed}} \ll X Q_b X^{-1/200+o(1)}, \tag{7}$$

and, jointly, its exact physical TT^* normalization supplies the packet target

$$|\mathcal{P}_{b,\mathrm{gen}}| \ll X^{1-\sigma_*+o(1)}. \tag{8}$$

The estimate is required only for the literal coefficient family generated by the TPC dictionary, not for arbitrary slopes, arbitrary weights, or a uniform Chowla family.

No derivation of (8) from (7) is asserted in this paper. The two displays are joint parts of H3. A future proof must exhibit the exact normalizing prefactor that turns the 1/200 quadratic saving into a 1/400 packet saving.

Hypothesis H3 is the parity-facing arithmetic gate. It is the boldest assumption in the paper and may contain most of the difficulty of the original problem. Calling it "spectral" does not make it a proved cancellation theorem.

Working Hypothesis 4.8 ([H] H5: determinant and distinguished zero mode)**.** *On the actual post-bin coefficient, there exist exponents $\lambda_D, \eta_Z \geq 0$ such that*

$$D_X \geq X^{-\lambda_D+o(1)} Q_X^3 J_X^2, \qquad |Z_X| \leq X^{-\eta_Z+o(1)} Q_X^2,$$

and

$$\lambda_D \leq 2\eta_Z.$$

The corresponding literal transfer inequality is assumed to introduce no further fixed-power loss beyond the exponents already charged in this determinant ledger and in Λ_{phys}. Ambient, random-model, pre-bin, or support-only energy is not a substitute for D_X, and a nonzero-frequency spectral gap is not a substitute for the estimate of Z_X.

H2, H3, and H5 constitute macro hypothesis **P**. H2 is a sufficient positive route for the resonance sector. If it fails, a signed filter-before-majorant replacement may be inserted, but the replacement must still deliver the packet bound at the same normalization.

4.3 Macro hypothesis S: strict physical endpoint

Working Hypothesis 4.9 ([H] H9: endpoint feasibility)**.** *After every frame, grouping, missing-high, localization, cover, boundary, and native-Gram cost has been charged exactly once, the physical synthesis operator obeys*

$$\left|\mathscr{S}_X\left(\{\mathcal{P}_b\}_{b\in\mathcal{B}_X^{\mathrm{ns}}}\right)\right| \leq X^{\Lambda_{\mathrm{phys}}+o(1)} \max_{b\in\mathcal{B}_X^{\mathrm{ns}}} |\mathcal{P}_b|,$$

with

$$\Lambda_{\mathrm{phys}} < \sigma_* = \frac{1}{400}.$$

Subpower block multiplicity is absorbed in the $o(1)$. Equality $\Lambda_{\mathrm{phys}} = 1/400$ does not close the route. The compatibility $\lambda_D = 2\eta_Z$ in H5 is not reusable as physical endpoint slack.

4.4 Strength and nonminimality

Remark 4.10 (The stack may be stronger than the target). The conjunction H1–H9 is designed to imply a scalar Hardy–Littlewood asymptotic, but the converse is neither claimed nor expected. A scalar asymptotic need not imply a positive bound on every fiber, a uniform cross-map majorant, or a canonical frame estimate. Thus "minimum viable" means a compact program exposing the interfaces currently identified in this series; it does not mean a logically minimal or independently easier reformulation of the twin-prime problem.

Remark 4.11 (The three resonance ledgers are not all remaining obstacles)*.* The quantities $W_X, \mathfrak{P}_X, \mathfrak{X}_X$ are the remaining ledgers in one positive resonant subcarrier. Generic signed arithmetic, full-block return, fixed-h_0 localization, determinant/zero-mode compatibility, high-frequency return, and the strict endpoint remain independent.

5 Conditional synthesis

The following implication is the only global theorem of TPC-MVP1. Its proof is short because the difficulty has been made explicit in the hypotheses.

Theorem 5.1 ([C] Conditional fixed-shift Hardy–Littlewood synthesis)**.** *Fix an admissible nonzero even integer h_0 and $W \in C_c^\infty((0,\infty))$. Assume H1–H9 for the literal TPC carrier, with every estimate taken in the same physical normalization and quantifier scope. Then*

$$\sum_{n\geq 1} \Lambda(n)\Lambda(n+h_0)W(n/X) = \mathfrak{S}(h_0)X \int_0^\infty W(t)\,dt + o(X). \tag{9}$$

Proof. Hypotheses H1, H4, and H6 produce an exhaustive subpower-size classification of physical blocks. Every term assigned to the soft remainder contributes $o(X)$ after the complete outer summation. The two established exceptional returns together with H2 handle the resonant blocks; H3 handles the generic signed blocks; and H4 returns every remaining short, high, ultra, boundary, and outer block. Thus every nonsoft raw block entering synthesis satisfies

$$\max_{b\in\mathcal{B}_X^{\mathrm{ns}}} |\mathcal{P}_b| \leq X^{1-\sigma_*+o(1)}.$$

Hypothesis H5 certifies that the determinant/zero-mode transfer used in that statement has no unrecorded incompatibility. H7 returns all auxiliary parameters to the prescribed h_0, and H9 gives

$$\left|\mathscr{S}_X\left(\{\mathcal{P}_b\}_{b\in\mathcal{B}_X^{\mathrm{ns}}}\right)\right| \leq X^{1-\sigma_*+\Lambda_{\mathrm{phys}}+o(1)} = o(X),$$

because $\Lambda_{\mathrm{phys}} < \sigma_*$. H8 identifies this synthesis output with $B_{h_0,\delta}(X)$, up to the already recorded $o(X)$ remainder. Therefore

$$B_{h_0,\delta}(X) = o(X).$$

Insert this in the established identity (2). □

Remark 5.2 (What the theorem does and does not prove). theorem 5.1 proves an implication from explicit unproved hypotheses to the target correlation. It does not prove H1–H9. In particular H3 contains the unresolved growing fixed-h_0 arithmetic cancellation, and the conjunction of the hypotheses may be stronger than (9).

5.1 The conditional prime-pair consequence

Corollary 5.3 ([C] Conditional fixed-gap prime pairs)**.** *Assume H1–H9 for a fixed admissible even h_0 and for one fixed nonnegative, nonzero $W \in C_c^\infty((1,2))$. Then*

$$\#\{p \in [X, 2X + O_{h_0}(1)] : p \text{ and } p+h_0 \text{ are prime}\} \gg_{h_0,W} \frac{X}{(\log X)^2}$$

for all sufficiently large X.

Proof. Since $\mathfrak{S}(h_0) > 0$ and $\int W > 0$, theorem 5.1 gives

$$\mathcal{C}_{h_0,W}(X) \gg_{h_0,W} X.$$

The terms for which n or $n+h_0$ is a prime power of exponent at least two contribute at most

$$O_{h_0,W}\left(X^{1/2}(\log X)^2\right) = o(X).$$

The remaining mass comes from prime-prime pairs, and each term is at most $O_W((\log X)^2)$. Division gives the stated lower bound. □

Corollary 5.4 ([C] Conditional $h_0 = 2$ specialization)**.** *If H1–H9 are eventually proved for the prescribed shift $h_0 = 2$, then corollary 5.3 implies infinitely many twin primes.*

Remark 5.5 (No sharp unsmoothing is needed for infinitude). A single fixed nonnegative smooth weight already yields the conditional lower bound in corollary 5.3. Smooth majorants and minorants may recover sharper interval formulations, but positivity and removal of higher prime powers are not the new parity-facing input. That input is H3, or an equally strong signed replacement for a failed positive route.

5.2 A proof skeleton in nine moves

The conditional theorem may be read as the following end-to-end program:

1. calibrate the singular-series main term and isolate $B_{h_0,\delta}$;
2. export the literal coefficient into canonical native packets;
3. return the two established exceptional resonance branches;
4. prove or replace the $W_X, \mathfrak{P}_X, \mathfrak{X}_X$ positive resonance gates;
5. prove signed generic affine Möbius cancellation on the actual carrier;
6. return every short, high, ultra, boundary, and outer sector;
7. verify determinant energy and the distinguished zero mode;
8. reassemble and localize to the prescribed h_0 with strict endpoint slack; and
9. reconnect to the original hard packet and use positivity only after the weighted asymptotic is obtained.

No move may be deleted by renaming its output.

6 Dependency graph and the role of dynamics

6.1 The full logical graph

The route is intentionally drawn with conjectural nodes visible.

[E] exact fixed-shift local-model identity (2)

$\Uparrow$

[C] original hard packet $B_{h_0,\delta}(X) = o(X)$

$\Uparrow$

[H] H8 literal reconnection $+$ **[H]** H9 strict endpoint

$\Uparrow$

[H] H6 full-block return $+$ **[H]** H7 fixed-h_0 localization

$\Uparrow$

[H] H1 canonical physical synthesis

$\Uparrow$

positive low-frequency resonance	generic/high sectors
[E] two exceptional returns	**[H]** H3 signed affine arithmetic
[E] three-ledger inequality	**[H]** H4 high/ultra return
[H] H2 $W/\mathfrak{P}/\mathfrak{X}$	literal outer coefficients

$\Downarrow$

[H] H5 determinant-energy/zero-mode compatibility

The graph has two important forks.

1. H2 is a *positive* sufficient route. If a literal principal or cross-map ledger is too large, the route may switch to a signed filter before taking absolute values. Failure of H2 does not disprove the scalar correlation.
2. H3 may be approached through direct native-h_0 determinant dispersion, a coherent prime-square theorem, or a signed affine large sieve. A result on an averaged shift or random coefficient is a different branch, not a proof of H3.

6.2 Where the parity difficulty lives

The program has already classified the local squarefree support in the primitive affine sector. After that classification the remaining sign is an actual growing Liouville/Möbius parity correlation [17]. Accordingly, the architecture does not claim that the parity barrier has been partly broken. It claims something more modest and testable:

> the literal carrier and normalization at which parity-sensitive cancellation must occur have been isolated more sharply.

H3 is the node where an actual arithmetic advance must enter. A support theorem, positive density, spectral decomposition, or finite certificate cannot substitute for it.

6.3 What the dynamical language contributes

The dynamical and spectral language used throughout the broader program contributes:

- canonical symbolic, orbit, conductor, and affine coordinates;
- transfer-like decompositions and exact finite resonance modes;
- Gram, projection, energy, holonomy, and width-sensitive norm identities;
- branch classification, provenance, and loss ledgers; and
- sharp countermodels that identify where a proposed transfer loses the physical zero mode.

These are genuine structural uses. In particular, the multiplicative character spectrum behind (3) replaces a crude worst-eigenvalue estimate by the exact filter factor $g_q(H)$ [5, 23].

6.4 What it does not contribute automatically

No dynamical or finite spectral identity in this program automatically supplies:

- independence of primes;
- Möbius randomness on the actual fixed-h_0 carrier;
- a fixed-shift arithmetic cancellation theorem;
- a breach of sieve parity; or
- an identification of Riemann zeros with eigenvalues of the TPC operators.

The safe summary is:

> *Dynamics organizes the arithmetic carrier and exposes where cancellation must occur; it does not itself supply that cancellation.*

No statement in TPC-MVP1 concerns the Riemann hypothesis, Hilbert–Pólya, or a zeta-spectrum realization.

6.5 Two exponent geometries

The current packet scales

$$Q_X = X^{267/400+o(1)}, \qquad J_X = X^{133/400+o(1)}$$

satisfy $Q_X J_X = X^{1+o(1)}$. On a resolved long fiber $N \geq J_X$, the established width-sensitive resonance norm replaces the crude exponent 267/800 by 67/400, a gain

$$\frac{267}{800} - \frac{67}{400} = \frac{133}{800}.$$

This is a real native spectral improvement on those fibers. It is not a global packet estimate because short fibers, principal mass, cross-map excess, and physical synthesis still have to be returned.

Independently, the physical synthesis has only the strict allowance

$$\Lambda_{\text{phys}} < \frac{1}{400}.$$

The large local numerical gain above may not be spent twice or transferred to unrelated blocks without a theorem.

7 Stop–go criteria and route switching

A route does not stop because a proof attempt is difficult. A genuine stop certificate is an actual-coefficient lower bound, counterexample, kernel, nonidentifiability theorem, or exponent obstruction in the claimed quantifier scope.

Gate	Go certificate	Genuine stop certificate	Response
Opened-atom cap	$W_X \le X^{o(1)}$ for actual normalized atoms	An actual family forces $W_X \ge X^{\omega}$ on active mass, with no compensating small total weight	Peel heavy atoms; refine conductor/length; analyze them with signs
Principal mass	$\mathfrak{P}_X \le X^{o(1)} Q_X^2$ in the literal inverse-length census	$\mathfrak{P}_X / Q_X^2 \ge X^{\pi}$ on actual packets	Stop the positive termwise bound; center or keep principal/nonprincipal cancellation
Map quotient	Identical affine maps aggregate with exact weight and recoverable provenance	Literal provenance forces polynomial unresolved multiplicity after canonical quotient	Publish collision classification; change the carrier
Cross-map excess	$\mathfrak{X}_X \le X^{o(1)} Q_X^2$ after identical maps are removed	Distinct actual maps force a fixed-power excess	Use signed filter before majorant, retaining phases and Möbius weights
Generic affine	(7) on the complete literal family	A coherent actual subfamily violates the required exponent, or the gate is proved equivalent to an unresolved stronger input	Change cutoff geometry, subtract a new main term, or publish the no-reduction theorem

Table 2: Local and arithmetic route decisions. Greek exponents in a stop certificate are fixed positive constants.

Gate	Go certificate	Genuine stop certificate	Response
Determinant / zero	$\lambda_D \le 2\eta_Z$ on the actual post-bin coefficient	An actual family forces $\lambda_D > 2\eta_Z$	Improve the zero mode, use a direct physical expansion, or publish the compatibility no-go
Canonical frame	Subpower condition cost on the inherited packet family	A genuine canonical condition number spends a fixed power at or beyond the budget	Switch to direct native-h_0 determinant dispersion
Localization	Auxiliary estimates return to the prescribed h_0 at recorded subcritical cost	Only averaged, density-one, or frozen-family control is possible	Publish the restricted theorem with its qualifier; do not promote it
Full return	Exact cover or separately controlled physical remainder	Actual-support remainder or native-Gram mismatch has fixed-power size	Repair the partition or publish the reassembly obstruction
Endpoint	$\Lambda_{\text{phys}} < 1/400$, with every loss charged once	$\Lambda_{\text{phys}} \ge 1/400$, including equality	Reoptimize parameters or publish endpoint infeasibility

Table 3: Physical return and endpoint decisions.

7.1 The positive-to-signed switch

The positive resonance strategy takes absolute values before the outer signed reconstruction. If its principal or cross-map ledger has a literal polynomial obstruction, the correct fallback is

$$\text{signed filter} \longrightarrow \text{arithmetic estimate} \longrightarrow \text{majorization},$$

not another paper on positive cross-cell dispersion. The signed object must retain:

- the reconstruction multiplier and its sign;
- the r-dependent unit phase;
- the affine character and actual moving row;
- the Möbius factors inside each resolved fiber; and
- coherent outer reassembly across frequency-free branches.

An estimate for independent r and branch variables does not control this product.

7.2 The three principal route families

1. **Positive resonance route.** Test W_X, then $\mathfrak{P}_X$, then the identical-map quotient, and only then $\mathfrak{X}_X$. This is the cheapest falsifiable route.
2. **Signed native route.** Attack H3 directly in the physical affine/determinant coordinates, bypassing a failed positive majorant or continuous Mellin lower frame.
3. **Coherent prime-square route.** Prove a coefficient-specific Bessel or dispersion estimate inside one residual coordinate, preserving the three nonsmooth coefficient slots and the physical prime weights.

The second and third routes may share exact dictionaries, but gains may be combined only when they act on disjoint costs or a theorem places them in one common carrier.

8 TPC-103 through TPC-120: a verification queue

The next sequence is a decision schedule, not a claim that eighteen papers remain before a proof. Each item may produce a positive theorem, a route-switching obstruction, or a sharper formulation of the next gate.

8.1 Why the inexpensive gates come first

TPC-103–106 concern positive or exact literal quantities. They can often be settled by factor bookkeeping, exact disintegration, or a counterexample before a deep arithmetic theorem is attempted. TPC-108 is deliberately delayed: proving a sophisticated affine Möbius estimate is wasteful if the positive normalization or the final synthesis already has a polynomial obstruction.

8.2 Promotion checkpoints

The queue has four formal checkpoints.

P1. After TPC-106, choose positive or signed resonance.

P2. After TPC-112, decide whether determinant transfer remains viable.

P3. After TPC-117, decide whether the literal full-block carrier exists with a controlled remainder.

P4. At TPC-118, stop unless the physical loss is strictly below 1/400. Only then may TPC-119 reconnect to the original scalar packet.

TPC-MVP2 is not scheduled as a proof announcement. It is a new audit that may conclude “go”, “reroute”, or “current endpoint infeasible”.

Paper	Working title	Decisive deliverable
103	Normalized opened-atom cap	Prove or refute $W_X \le X^{o(1)}$ from the literal factor list.
104	Resolved principal-mass disintegration	Open $\mathfrak{P}_X$ before width grouping; return $N = 1, 2$ exactly.
105	Provenance-preserving affine-map quotient	Aggregate identical maps without losing source recovery or weight.
106	Width-weighted distinct-map incidence	Test the positive $\mathfrak{X}_X$ target on genuine distinct maps.
107	Signed filter before positive majorization	Construct the exact signed replacement if TPC-104 or 106 stops.
108	Literal generic-affine Möbius dispersion	Attempt the restricted $X^{-1/200}\, TT^*$ gain in H3.
109	Coherent prime-square block test	Seek a coefficient-specific saving or a saturation theorem on one actual residual block.
110	Post-bin determinant energy on actual support	Prove a natural-scale lower bound or an actual energy obstruction.
111	Distinguished zero-mode cancellation	Estimate Z_X independently of the energy census.
112	Determinant–zero compatibility audit	Test $\lambda_D \le 2\eta_Z$ with no double charging.
113	Canonical physical frame and reassembly	Measure the inherited synthesis cost or prove a condition-number lower bound.
114	Residual grouping and native-Gram identification	Control $s = d(m)r$ and prove the exact Gram intertwining.
115	Fixed-h_0 de-aliasing and localization	Return frozen and frequency-averaged estimates to one prescribed shift.
116	High-frequency and ultra-tail closure	Return the complete complement at the original normalization.
117	TPC-18 full-block physical remainder	Prove exact cover or estimate the physical remainder.
118	Strict endpoint optimization	Assemble a nonduplicated loss ledger and require $< 1/400$.
119	Original hard-packet reconnection	Audit H8 against the TPC-15 scalar packet.
120 / MVP2	Global route decision	Insert proved gates, declare failed routes, and decide whether a conditional closure implementation is justified.

Table 4: A falsifiable verification queue after TPC-MVP1.

9 Publication-worthy negative outcomes

The architecture is useful even if a main route fails, provided the failure is mathematical rather than autobiographical. The following would be legitimate standalone outcomes when proved on actual coefficients:

1. a polynomial opened-atom concentration theorem;
2. a sharp principal-mass obstruction to the positive majorant;
3. a literal example showing that positive majorization destroys a signed cancellation present before the filter;
4. a canonical frame condition-number lower bound;
5. an equivalence or no-reduction theorem between the generic affine gate and a prescribed-shift Chowla-type input;
6. determinant energy and zero-mode incompatibility on an actual family;
7. an average-to-fixed-h_0 localization obstruction;

8. an actual-support remainder obstruction to full-block cover;
9. a proof that the present parameter geometry forces $\Lambda_{\mathrm{phys}} \geq 1/400$; or
10. nonidentifiability of the required physical Gram from the available surrogate Gram.

Remark 9.1 (Failure to prove is not a theorem)*.* Computational saturation, a stalled argument, or the absence of a known estimate is not a stop certificate. A negative paper requires an exact family, lower bound, kernel, counterexample, or logical nonimplication with the physical typing visible.

9.1 Weaker positive outputs

If a fixed-h_0 gate remains open, the same methods may yield valuable but strictly qualified results:

- averaged-shift or density-one-shift cancellation;
- a fixed-power gain on a literal selected block;
- a density-one cofactor range;
- a theorem after smoothing or freezing one coefficient slot;
- a complete classification of saturating resonance families; or
- a parameter range in which the endpoint is feasible.

These results may be publishable. They remain L0/L1 or a restricted arithmetic theorem unless a fixed-h_0 return is proved.

9.2 A portfolio rather than a single bet

The MVP program therefore has three kinds of success:

(a) a gate closes and the dependency graph advances;

(b) a gate fails by a sharp theorem and an entire false route is retired; or

(c) the literal audit reveals a better carrier on which a new route can be formulated.

What it does not permit is indefinitely renaming the same unresolved arithmetic cancellation.

10 Finite route audit

The standard-library script `experiments/tpc_mvp1_route_audit.py` encodes the dependency graph used in this paper. It audits 98 recorded items:

- 19 nodes are typed as established, hypothetical, or conditional;
- all 18 declared dependency edges have valid endpoints and the graph is acyclic;
- all nine hypotheses are ancestors of the conditional Hardy–Littlewood node;
- no established node is labeled L2;
- all ten route gates contain nonempty go, stop, and fallback records;
- the TPC-103–120 queue is consecutive and has 18 entries; and
- five rational identities in the scale and endpoint ledgers are exact.

The committed output is `experiments/tpc_mvp1_route_audit.json`.

In particular the audit checks

$$\frac{267}{400}+\frac{133}{400}=1, \qquad \frac{267}{800}-\frac{67}{400}=\frac{133}{800},$$

$$\frac{1}{200}=2\cdot\frac{1}{400}, \qquad \frac{1}{400}-\frac{1}{500}=\frac{1}{2000}>0.$$

The sample $1/500$ is only a consistency witness for strict endpoint slack, not a proved physical loss.

Remark 10.1 (Certificate scope). The route audit verifies graph syntax, claim typing, and rational bookkeeping. It does not test H1–H9, asymptotic Möbius cancellation, prime-pair data, or the plausibility of a parity breakthrough. A passing finite audit is L0 evidence only.

11 Claim boundary and final verdict

[E] Established in the imported series

- The fixed-shift target has an exact hard-packet anchor.
- Important sectors possess literal native dictionaries and exact determinant/resonance decompositions.
- Two exceptional literal resonance branches are returned at their recorded endpoint scale within the present low-window decomposition and native normalization.
- The remaining positive resonant carrier satisfies the three-ledger inequality (3).
- Several false transfers and coefficient-blind shortcuts have rigorous countermodels.
- Full-block return, generic arithmetic dispersion, determinant/zero-mode compatibility, and the physical endpoint are separately required gates.

[H] Explicitly assumed here

H1–H9 are working hypotheses. In particular, no established result currently supplies the growing fixed-h_0 signed affine Möbius estimate H3, the complete physical return H6, or the full strict endpoint statement H9.

[C] Proved conditionally

If H1–H9 all hold in one literal normalization, then the hard packet is $o(X)$, the weighted Hardy–Littlewood asymptotic follows, and a nonnegative weight gives a fixed-gap prime-pair lower bound. The specialization $h_0 = 2$ is a conditional corollary only.

[N] Not established

TPC-MVP1 does not establish:

- any new L2 growing fixed-h_0 arithmetic estimate;
- H1–H9, individually or jointly;
- that the hypothesis stack is weaker than, or independently easier than, the original hard packet;
- a sieve-parity breakthrough, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime theorem;
- a Riemann-hypothesis consequence or a zeta-zero spectral realization; or
- a numerical measure of how close the program is to a proof.

Current honest verdict

After 102 technical papers, the program has enough verified structure to state an end-to-end candidate conditional route with all currently identified edges shown. That is a meaningful change: the maze now has named doors, dependency arrows, kill criteria, and fallback corridors. It is not the same as having opened the parity-facing door.

The immediate rational strategy is therefore:

$$\begin{aligned}&\text{cheap literal kill tests}\\ \Longrightarrow\ &\text{positive or signed resonance decision}\\ \Longrightarrow\ &\text{first genuine fixed-}h_0\text{ arithmetic attack}\\ \Longrightarrow\ &\text{physical return and endpoint audit.}\end{aligned}$$

If any actual-coefficient obstruction closes a corridor, that negative theorem is part of the program's success. If the corridor remains open, the next paper must prove a quantitative statement at its own declared level; it may not substitute another representation for the same unresolved cancellation.

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